12.5 Scalability solutions

Many solutions have been proposed, and quite a few of them have been in operation in production networks. The Ethereum community is working hard to address the scalability issues. The proposed solutions are across all levels of the stack.

12.5.1 Side channel

Side channel is a solution at the blockchain application level. The state channel of Ethereum and the lightning channel of Bitcoin are examples. The idea is to keep only relevant transactions on-chain for confirmation and recording. Other transactions between trusted parties are offloaded to a side channel, thus reducing the transaction load on the main channel. Periodically, the gist of the off-chain happenings are synchronized with a transaction on the main channel. The transactions of off-chain channels take place at much higher speeds than on the blockchain networks because no consensus or recording on the blockchain distributed ledger is required.

Recall that you used a side channel in the MPC-Dapp in chapter 7. This Dapp addresses scalability to a certain extent at the application level. At this point, you have the knowledge and skill to apply the side channel concept when developing Dapps, as demonstrated in MPC-Dapp. Try using the side channel model where it is relevant in your Dapp development.

12.5.2 Block size

Increasing block size is a protocol-level solution. Transaction time depends on block time. So why not increase the size of the block